Reg. No. : | | | | | | | | | | | | |

Question Paper Code : 40994

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2021.

Seventh Semester

Aeronautical Engineering

OCE 751 — ENVIRONMENTAL AND SOCIAL IMPACT ASSESSMENT

(Common to Aerospace Engineering/Agriculture Engineering/Automobile Engineering/Biomedical Engineering/Computer Science and Engineering/Computer and communication Engineering/Electronics and communication Engineering/Electronics and Telecommunication Engineering/Environmental Engineering/Geoinformatics Engineering/Industrial Engineering/Industrial Engineering and Management/Manufacturing Engineering/Marine Engineering/Material Science and Engineering/Mechanical Engineering/Mechanical Engineering (Sandwich)/Mechanical and Automation Engineering/Mechatronics Engineering/Medical Electronics/Petrotechnical Engineering/Production Engineering/Robotics and Automation/Bio Technology/Chemical Engineering/Chemical and Electrochemical Engineering/Fashion Technology/Food Technology/Food Science and Technology/ Food Technology and Management/ Technology/Petrochemical Technology/Petroleum Engineering/Pharmaceutical Technology/Textile Chemistry/Textile Technology)

(Regulations 2017)

Time : Three hours | Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Define EIA.

2. State any two types of EIA.

3. What do you understand by Environmental Inventory?

4. Differentiate between reversible and irreversible impacts.

5. What is Life cycle assessment?

6. Define rehabilitation process in EMP.

7. State the benefits of cost benefit analysis.

8. Define exposure assessment.

9. State any two impacts of thermal power plant projects.

10. What are ways to mitigate noise due to airports projects?

PART B — (5 × 13 = 65 marks)

11. (a) Explain the different elements of EIA in relation to the Project Cycle and Briefly describe the legal provisions of EIA in India.

Or

(b) Describe the various steps involved in planning the baseline data collection for EIA.

12. (a) Draw a flow diagram of showing different phases of impact assessment and explain them in detail.

Or

(b) Describe the checklist for EIS preparation of impact on land.

13. (a) Discuss the important issues involved in developing an environmental monitoring programme and explain the same.

Or

(b) Explain the role of public participation in EIA process in various developmental projects with some examples.

14. (a) Write a detailed note on " Resettlement" and "Rehabilitation" policy.

Or

(b) Discuss the social economic impacts of "Bharatmala Project" and Write a note on cost benefit analysis.

15. (a) List out the adverse impacts of a "Irrigation Project" and suggest appropriate mitigation measures for each.

Or

(b) How would you prepare the EIA report in order for road and bridge projects?

PART – C (1 × 15 = 15 marks)

16. (a) Discuss the mechanism by which the Environmental Impact Assessment (EIA) process could be applied to government and federal policies and programmes.

OR

(b) Discuss the various environmental impacts of MRTS (Mass Rapid Transit System) in India and prepare the EIA process for MRTS projects.